

Test Case Report for Guest Identity, Reservation & Digital Key Generation Smart Contract

Test No	User Story (Redeemer)	Conditions	Steps	Expected Output	Accepted Output	Result
1	Initial Submit-Valid	All fields in datum are empty (identity, reservation, key Info, guestAddress)	Submit guest address and adminPKH	GuestAddress and AdminPKH are set correctly and other fields remain unchanged	Update accepted & committed	✅ Pass
2	Initial Submit – Reject	guestAddress is already set	Attempt to submit identity again	Validator rejects due to existing identity submission	Update rejected	✅ Pass
3	FullReservation – Valid	Guest address and adminPKH is set and reservation fields are empty	Submit reservation details (room ID, dates, etc.)	Reservation fields updated; identity, key, admin unchanged	Update accepted & committed	✅ Pass
4	FullReservation – Reject	Reservation already exists	Try to reserve again	Validator rejects update to prevent overwriting existing	Update rejected	✅ Pass

				reservation		
5	FullIdentitySubmit – Valid	Reservation exists, identity is empty	Submit full identity fields (name, pass, photo)	Identity fields updated; reservation, key, guestAddress unchanged	Update accepted & committed	✓ Pass
6	FullIdentitySubmit – Reject	Identity already present	Attempt to re-submit identity	Validator rejects duplicate submission	Update rejected	✓ Pass
7	GenerateAndValidateKey – Valid	Identity & reservation confirmed, keyInfo is empty	Generate and validate digital key	keyInfo set with key and validation flags; other fields unchanged	Update accepted & committed	✓ Pass
8	GenerateAndValidateKey – Reject	Key already present	Attempt to regenerate key	Validator rejects update	Update rejected	✓ Pass
9	CheckOut – Valid	All previous steps completed (identity, reservation, key validated)	Call CheckOut	reservation and keyInfo cleared; identity, guestAddress, admin remain intact	Update accepted & committed	✓ Pass
10	CheckOut – Reject	Reservation or keyInfo not set	Call CheckOut prematurely	Validator rejects due to missing	Update rejected	✓ Pass

				preconditions		
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Building test suite 'full-process-test' for smart-contract-1.0.0...
[1 of 1] Compiling Main ( Test/NewGuestTest.hs, /home/prithwish-chakraborty/Documents/Codes/NewHGE/HGE_SC/dist-newstyle/build/x86_64-linux/ghc-8.10.7/smart-contract-1.0.0/t/full-process-test/build/full-process-test/full-process-test-tmp/Main.o )
Linking /home/prithwish-chakraborty/Documents/Codes/NewHGE/HGE_SC/dist-newstyle/build/x86_64-linux/ghc-8.10.7/smart-contract-1.0.0/t/full-process-test/build/full-process-test/full-process-test ...
Running 1 test suites...
Test suite full-process-test: RUNNING...
Guest Validator Tests
  Unit Tests
    FullReservation: valid update changes only reservation: OK
    FullIdentitySubmit: changes only identity: OK
    GenerateAndValidateKey: only keyInfo changed correctly: OK
    CheckOut: clears reservation and keyInfo only: OK
    Invalid: FullReservation also modifies identity (should not happen): OK
    Invalid: FullIdentitySubmit without reservation (violates precondition): OK
    Invalid: GenerateAndValidateKey with unverified identity: OK
    Invalid: CheckOut wipes identity (should not happen): OK
  Property-Based Tests
    FullReservation: only reservation updated from initial state: OK
      +++ OK, passed 100 tests.
    FullIdentitySubmit: only identity updated after reservation: OK
      +++ OK, passed 100 tests.
    GenerateAndValidateKey: only keyInfo updated after identity & reservation: OK
      +++ OK, passed 100 tests.
    CheckOut: clears reservation and keyInfo, keeps rest unchanged: OK
      +++ OK, passed 100 tests.

All 12 tests passed (0.00s)
Test suite full-process-test: PASS
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